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PERMANENT OR TEMPORARY BIOPSY CLIPS?

Insights from a Single-Institution Survey of Singapore Women's Preference for Breast Biopsy Marker Types



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BACKGROUND

Metallic clip markers are commonly deployed following breast biopsies for lesion localisation.

As these clips may be left in situ indefinitely, some women feel uneasy with retaining a permanent foreign body and may even decline clip placement.

OBJECTIVES

To evaluate women's attitudes towards permanent metallic clip markers and whether hypothetical biodegradable clip markers would be a more acceptable alternative.

METHODS

A prospective survey of 162 women was conducted from 2015-2024.

Women aged > 21 years old were recruited from three groups:
(1) patients undergoing breast needle biopsy requiring clip placement
(2) women with previous breast biopsy clip insertion, and
(3) control group comprising female breast care doctors (including surgeons, radiologists and pathologists).

Participants completed a questionnaire on their preference for a permanent metallic or hypothetical biodegradable clip, as well as reasons underlying their choice.

RESULTS

Demographics

Total sample size (n = 162)

43.8% Patients undergoing breast biopsy (n = 71)

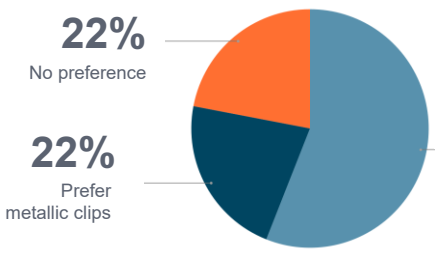
45.1% Patients with previous clip placement (n = 73)

11.1% Female breast care doctors (n = 18)

Results

56%
A majority of respondents preferred biodegradable breast biopsy clips

84.6%
Of women who chose biodegradable clips cited the psychological unease and possible safety issues with the permanent retention of a foreign body



Breakdown of patient preferences for breast clip material

Specific groups that preferred biodegradable clips:

60% Women with clips or undergoing biopsy **vs** **28%** Female breast care doctors (p = 0.012)

73% Women undergoing a biopsy **vs** **47%** Women with existing clips (p = 0.001)

Findings

Patients > doctors preferred biodegradable clips, because they are less familiar with metallic clip safety.

Women with **existing clips** were **less likely to prefer biodegradable ones**, possibly because they had become accustomed to the metal clip over time.

CONCLUSION & IMPLICATIONS

Main Findings

Majority of women **prefer biodegradable** breast biopsy clip over metallic one.

Most popular reasons **being psychological unease** and potential **safety issues** with the permanent retention of a foreign body.

Clinical Impact

These concerns **influence patients' acceptance of clip deployment** and the overall biopsy **experience**.

Future Directions

Survey highlights the need to **develop alternative, non-permanent clips** to address patients' concerns and improve breast biopsy experience.

REFERENCES

1. Lee IT, Ma KS, Luan YZ, Chen JL. Immediate clip migration after breast biopsy: a meta-analysis for potential risk factors. Br J Radiol. 2022;95(1140):20220195. doi:10.1259/bjr.20220195

2. Zada MH, Zehava Gallimidi, Schlesinger−Laufer M, Nyska A, Domb AJ. Biodegradable Breast Tissue Marker Clip. ACS Applied Bio Materials2020;3(11):7439-7453. doi:10.1021/acsabm.0c00655